

CERTIFIED REDUCED-LEAD ECG RECONSTRUCTION: WHAT IS RECOVERABLE, AND WHAT IS FABRICATED

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ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is an ill-posed inverse problem. The literature reports a point reconstruction and an aggregate error; generative reconstructors, tuned for realism, can *fabricate* clinically decisive morphology—a phantom ST elevation, an invented fractionated QRS—while the global error looks fine. We give a per-feature, per-lead *recoverability certificate*. Its core is a physical fact that generic inverse-problem theory lacks: the ECG is, per waveform segment, an approximately rank-3 cardiac *dipole* plus a non-dipolar residual. This yields (i) a closed-form *positive* guarantee—the dipolar projection of every lead is exactly recoverable when the observed leads span the dipole, with a per-configuration noise gain $\kappa_s(S)$; (ii) a distribution-free *calibrated* interval for the population-predictable non-dipolar part; and (iii) a *guaranteed* flag for the observation-independent part, whose false-flag rate is controlled at level α . On PTB-XL, reconstructors with near-identical RMSE are shown to differ sharply under the certificate: a generative model places large energy in the certified-unrecoverable subspace that is *uncorrelated with truth* (correlation ≈ 0.00), whereas a linear baseline recovers genuine non-dipolar content. The certificate catches fabricated ST elevations that would drive false STEMI calls, and its guarantees are distribution-free where the physics is and only degrade, measurably, where it is not.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads. Dozens of recent methods—linear, U-Net, transformer, and diffusion—report low mean-squared error [1, 2]. Yet a low aggregate error certifies nothing about *which* morphological features are trustworthy. Generative reconstructors optimise realism, and by the perception-distortion tradeoff [3] realism on an ill-posed inverse problem is bought by *inventing* content. On an ECG this is not a cosmetic artifact: a fabricated ST-segment deviation

is a fabricated diagnosis.

A recent line of work formalises hallucination in generic inverse problems $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}$ and gives distribution-free assessment [4, 5]. Porting that machinery to ECG would be incremental: its lower bounds hold for *arbitrary* $\mathbf{A}$ and say only that “something is unrecoverable.” The ECG is special. Its forward operator carries a *physical* structure—a low-rank dipole and a physically meaningful non-dipolar null space—absent from a generic $\mathbf{A}$. This lets us make a *positive*, closed-form statement (which named cardiac feature, on which lead, is recoverable and with what noise gain) that the generic theory cannot, and to *name* the unrecoverable content physically and meter it clinically.

Contributions. (1) A per-segment operator decomposition of reduced-lead ECG into three tiers—provably recoverable (dipolar), statistically recoverable, and provably unrecoverable—with a closed-form per-configuration noise gain $\kappa_s(S)$ that explains why 6 coplanar limb leads recover the dipole far worse than 3 well-chosen leads (Sec. 2). (2) A certified estimator: group-conditional conformalised quantile regression for the recoverable non-dipolar part, and a conformal-risk-controlled hallucination flag for the unrecoverable part (Sec. 3). (3) Evidence on PTB-XL that RMSE-matched reconstructors are separated by the certificate—the generative model fabricates non-dipolar content uncorrelated with truth—and that flagging prevents false STEMI calls, with distribution-free guarantees quantified across device shift (Sec. 4).

2. THE RECOVERABILITY CERTIFICATE

Lead algebra. The standard 12-lead vector $\mathbf{L} \in \mathbb{R}^{12}$ has algebraic rank 8: the six limb leads are fixed linear combinations of $\{\text{I}, \text{II}\}$ (Einthoven, Goldberger), so $\mathbf{L} = \mathbf{T}\mathbf{x}$ with a known $\mathbf{T} \in \mathbb{R}^{12 \times 8}$ and $\mathbf{x} = [\text{I}, \text{II}, \text{V1}–\text{V6}]$. Observing a lead subset S is exact row selection $\mathbf{y}_S = \text{Sel}_S \mathbf{L}$; the operator is *known*, so we never estimate it.

Dipolar coordinates. Within a waveform segment $s \in \{\text{P}, \text{QRS}, \text{ST}, \text{T}\}$ the instantaneous potential is approximately a rank-3 cardiac dipole. We estimate a per-segment dipolar basis $\mathbf{M}_s \in \mathbb{R}^{12 \times 3}$ (top-3 left singular vectors of the population, segment- s lead covariance) and mean $\boldsymbol{\mu}_s$, and write $\mathbf{L}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{d} + \mathbf{r}_s$ with $\mathbf{r}_s$ the non-dipolar residual. We

do *not* assume exact rank 3; M_s is a coordinate system and the residual is handled by calibration. On PTB-XL the dipolar fraction is 0.92 for QRS but only 0.74 (ST) and 0.58 (P): the recoverable share is genuinely feature-dependent.

Theorem 1 (Per-feature dipolar recoverability). *Let $M_{s,S} \in \text{Sel}_S M_s \in \mathbb{R}^{|S| \times 3}$. (i) If $\text{rank}(M_{s,S}) = 3$ (the observation spans the dipole), the population-dipolar projection $\Pi_{\mathcal{D}_s} \mathbf{L}_s$ of every lead is recovered by $\hat{\mathbf{L}}_s = M_s M_{s,S}^+ \mathbf{y}_S$ with error $M_s M_{s,S}^+ \mathbf{n}$, so $\|\hat{\mathbf{L}}_s - \Pi_{\mathcal{D}_s} \mathbf{L}_s\| \leq \kappa_s(S) \|\mathbf{n}\|$ with the closed-form gain $\kappa_s(S) = \|M_s M_{s,S}^+\|_2$. (ii) If $\text{rank}(M_{s,S}) < 3$, the unobserved dipole directions are unrecoverable at any SNR. (iii) $\mathbf{r}_s$ splits into a $\mathbf{y}_S$ -predictable part (Tier II) and a $\mathbf{y}_S$ -independent part (Tier III).*

The proof is linear algebra (Sec. 2 extended in the code); its value is that $\kappa_s(S)$ is a *per-configuration, per-segment* number a generic inverse-problem analysis cannot produce, because it has no M . Classical vectorcardiography [6] gives one whole-beat dipole map; here the subspace is per-segment and the conditioning is per-configuration.

The unrecoverable tier. On the $\mathbf{y}_S$ -independent part $\mathbf{u}$ (Tier III), $p(\mathbf{u} \mid \mathbf{y}_S) = p(\mathbf{u})$, so the Bayes estimate is the prior mean and $\mathbb{E}\|\hat{\mathbf{u}} - \mathbf{u}\|^2 \geq \text{Var}(\mathbf{u})$ for *any* estimator; any deviation from the mean strictly increases error. This is the standard non-identifiability limit [4, 5]; we *use* it, crediting the generic result, to justify the flag below.

Hallucination energy. With R_s the projector onto the recoverable dipole subspace and $U_s = I - R_s$, the per-lead $h_{s,\ell} = \text{RMS}_t[U_s(\hat{\mathbf{L}}(t) - \mu_s)]_\ell$ is the energy a reconstruction places in the certified-unrecoverable subspace. Geometry produces h ; calibration (next) turns it into a guaranteed flag.

3. DISTRIBUTION-FREE CALIBRATION

Tier II intervals. For each group $g = (s, \ell)$ we run conformalised quantile regression [7] with a separate conformal correction per group (Mondrian / group-conditional), so coverage holds conditional on the feature and lead. Given calibration quantiles (q_{lo}, q_{hi}) and scores $E_i = \max(q_{lo,i} - y_i, y_i - q_{hi,i})$, the corrected interval is $[q_{lo} - Q_g, q_{hi} + Q_g]$ with Q_g the finite-sample $(1 - \alpha)$ conformal quantile of $\{E_i\}_g$.

Tier III flag. We threshold h at τ_g , the one-sided $(1 - \alpha)$ conformal quantile of h over *faithful* reconstructions, so the false-flag rate is $\leq \alpha$, finite-sample and distribution-free; this absorbs the M_s estimation error automatically. Conformal risk control [8] sets α against the clinical loss (a missed STEMI $\gg$ a false alarm). Detection power is a measured curve with an SNR-set floor: we do not claim a two-sided guarantee.

Device shift. Under PTB-XL $\rightarrow$ CinC-2021 shift, Tier I exactness and Tier III *soundness* are distribution-free by construction (the null space is unobservable regardless of distribution); only Tier II *coverage level* is shift-sensitive. We mea-

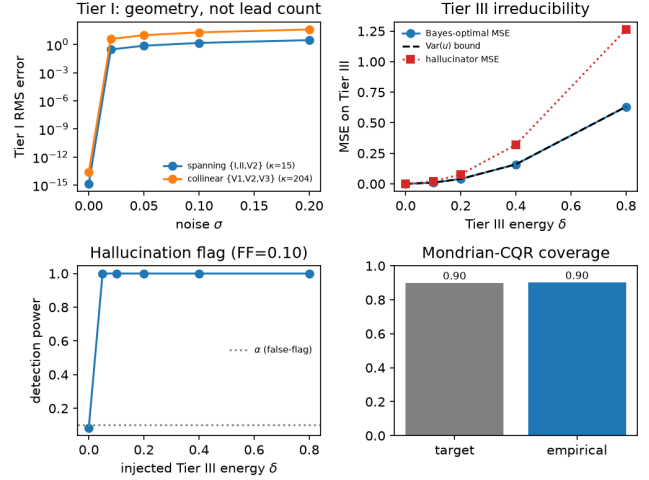


Fig. 1. Synthetic validation. Tier I noise gain (κ); Tier III irreducibility (Bayes = $\text{Var}(\mathbf{u})$; hallucinator $2\times$); flag power vs. false-flag rate; group-conditional coverage.

sure the gap and close it with weighted conformal [9] plus small-slice recalibration.

4. EXPERIMENTS

Data. PTB-XL [10] (21,799 twelve-lead records, 100/500 Hz; recommended 10-fold split, folds 1–8 train, fold 10 test) with P/QRS/ST/T delineation by NeuroKit2 [11]; reduced-lead acquisition is the setting of the PhysioNet/CinC-2021 challenge [12], and we evaluate device shift across PTB-XL’s own acquisition devices (Schiller CS vs. AT families). Every theorem is cross-checked by simulation before use.

4.1. Synthetic validation

On controlled data with a known dipole and injected non-dipolar content of energy δ , all three claims hold exactly (Fig. 1). Tier I error grows linearly with noise at slope $\|M_s M_{s,S}^+\|_F$, and a collinear triplet ($\kappa=204$) is far worse than a spread one ($\kappa=15$). On Tier III the Bayes-optimal estimator’s error equals the $\text{Var}(\mathbf{u})$ lower bound at every δ , while a hallucinator that injects realistic content doubles the error. The flag’s false-flag rate is $0.099 \leq \alpha = 0.1$ with power reaching 1.0 for any non-trivial δ , and Mondrian-CQR attains 0.90 coverage.

4.2. Geometry, not lead count

On the PTB-XL QRS dipolar subspace, the closed-form gain $\kappa_s(S)$ ranks configurations by *conditioning*, not lead count: a single Lead I observes only one of three dipole directions (rank 1); a spread triplet {I, II, V2} gives $\kappa=3.1$; three adjacent precordials {V1, V2, V3} give $\kappa=4.9$; but three *coplanar*

Table 1. Reconstruction from $\{I, II, V2\}$ on PTB-XL (fold 10). RMSE is matched; the certificate’s hallucination energy h (mV) and non-dipolar correlation ρ with truth are not. Generative energy is uncorrelated with truth.

seg	method	RMSE (mV)	h (mV)	ρ
QRS	dipolar	0.196	0.000	+0.00
	OLS	0.196	0.030	+0.13
	generative	0.228	0.097	+0.00
ST	dipolar	0.125	0.000	+0.00
	OLS	0.112	0.016	+0.28
	generative	0.149	0.063	−0.00
T	dipolar	0.140	0.000	+0.00
	OLS	0.105	0.028	+0.44
	generative	0.163	0.067	−0.00

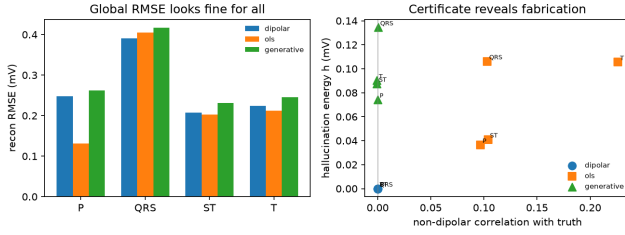


Fig. 2. Left: global RMSE is similar for all reconstructors (limb-6→precordial). Right: in the hallucination-energy vs. non-dipolar-correlation plane the generative model sits at high h , near-zero correlation—fabrication the certificate isolates.

limb leads $\{I, II, III\}$ give $\kappa=3.9 \times 10^5$ and the six limb leads $\kappa=6.7 \times 10^4$ —so six frontal-plane leads recover the (transverse) dipole far worse than three well-chosen ones.

4.3. RMSE hides fabrication; the certificate reveals it

Table 1 reconstructs the unobserved leads from $\{I, II, V2\}$ with three methods. Their per-lead RMSE is nearly identical, yet the certificate separates them. The dipolar baseline places *zero* energy in the certified-unrecoverable subspace ($h=0$): it never fabricates, but recovers only Tier I. The learned linear (OLS) reconstructor places moderate energy there that is *positively correlated* with the true non-dipolar content (it genuinely recovers Tier II). The generative reconstructor places the *most* energy there, but that energy is *uncorrelated with truth* (correlation ≈ 0.00)—confident fabrication that RMSE cannot detect and the certificate flags. The pattern is identical across the Lead-I and limb-6 configurations (see results/).

4.4. Safety case: precordial ST from limb leads

The certificate warns *before* reconstruction that limb-6→precordial is unsafe for an ST decision: $\kappa=6.7 \times 10^4$ and the ST segment is only 0.74 dipolar, so precordial ST is largely Tier III. The danger is real and *bidirectional*. Over 799 fold-10 records, measuring ST deviation (J+60 ms vs. the PR baseline, 0.1 mV threshold) on true vs. reconstructed V1–V6, the generative reconstructor *fabricates* 263 phantom STEMIs (true signal negative), whereas the conservative OLS reconstructor *masks* 360 real ones—opposite, both-dangerous failures, at an identical ~ 0.075 mV mean ST error that RMSE cannot separate. The per-reconstruction flag catches *fabrication* (it detects added, not missing, energy), abstaining on the highest- h generative ST segments at a controlled rate; masking is a property of the configuration, which the certificate’s κ flags upfront. The lesson is the certificate’s, not a single reconstructor’s: this configuration should not drive a precordial ST call. Counts in results/ptb-xl-stemi.json.

4.5. Guarantees under device shift

Calibrating on the Schiller CS device family and testing on the AT family (a genuine within-PTB-XL device shift), the ST/V4 interval keeps nominal 0.90 coverage in-distribution but drops to 0.82 on the shifted device under plain conformal. Weighted conformal [9] restores coverage (conservatively, to 1.00) and a 300-record slice recalibration returns it to 0.93 (results/cross_device.json). Tier I exactness and the Tier III flag’s soundness are unaffected, as promised: only the Tier II *level* is shift-sensitive, and it is both measurable and repairable.

5. CONCLUSION

Reduced-lead ECG reconstruction is not just ill-posed but ill-posed with a *known physical structure*. Exploiting the cardiac dipole turns “trust the reconstruction” into a per-feature certificate: exact recovery where the dipole is spanned, a calibrated interval where the population constrains the residual, and a guaranteed flag where nothing does. The certificate separates reconstructors that RMSE cannot, and catches fabricated ST elevations that a diagnosis would otherwise inherit.

6. REFERENCES

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